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RESEARCH PAPER

Enhancing e-commerce recommendations: A mechanism to detect and mitigate spam reviews

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Abstract

The prevalence of spam reviews on e-commerce platforms undermines consumer trust, distorts product evaluations, and often leads to suboptimal purchasing decisions. This study proposes a spam-aware recommendation framework that first detects spam reviews using machine learning and deep learning models based on semantic, sentiment, and metadata features, and then integrates these filtered credible reviews into personalized product recommendations. Rather than treating spam detection and recommender design as isolated tasks, this work bridges the gap between review credibility and user preference modelling to improve recommendation accuracy. By ensuring that only trustworthy reviews contribute to recommendation generation, the proposed approach enables users to receive more reliable and preference-aligned suggestions, thereby supporting more informed and confident purchasing decisions.

Keywords Fake review detection · Spammer detection · Semantic analysis · Sentimental analysis · Recommendation mechanism

JEL Classification M31 · D83 · C55

Introduction

The digital revolution has irrevocably transformed global retail, ushering in an era in which e-commerce has become a dominant mode of consumer exchange. Projections estimate that the global e-commerce market will reach approximately US$4.32 trillion by 2025 (Statista, 2025), with continued growth toward 3.6 billion online shoppers by 2029. Platforms such as Amazon now account for more than 40% of U.S. online retail sales (eMarketer, 2024), demonstrating the central role of platform-mediated digital markets in shaping contemporary consumption.

Despite this unprecedented scale, digital marketplaces face a fundamental challenge of information asymmetry. Online reviews are intended to reduce uncertainty by conveying peer-generated information about product quality. Indeed, nearly 95% of consumers report relying on online reviews prior to purchase (Brightlocal, 2025). However, many products on large platforms accumulate thousands of reviews, far exceeding consumers' ability to meaningfully process the information. This review overload imposes substantial cognitive burden (Furner & Zinko, 2017; Hu & Krishen, 2019) and heightens concerns regarding the credibility of review content.

A central driver of this credibility concern is the presence of deceptive review content. Prior research distinguishes among incentivized reviews, fake reviews, and coordinated opinion spam (Mayzlin et al., 2014; Z. Wu et al., 2024). Importantly, incentivized reviews—defined as feedback provided in exchange for monetary or non-monetary rewards—are not inherently deceptive, whereas fake reviews are intentionally fabricated to mislead consumers. In this study, we adopt the broader concept of spam reviews as an umbrella

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construct, encompassing deceptive reviews generated through coordinated human manipulation, incentive-driven deception, and fabricated content. This conceptualization allows us to distinguish spam reviews from benign incentivized feedback while maintaining terminological consistency throughout the manuscript. Against this background, this study focuses on detecting spam reviews and, crucially, on how identified spam can be systematically integrated into product recommendation systems to improve personalization quality.

Empirical evidence suggests that deceptive reviews constitute a substantial and systematic problem in online marketplaces. Independent audits and platform investigations report that a significant proportion of online reviews may be misleading or strategically manipulated, particularly in highly competitive product categories (He et al., 2022; Mayzlin et al., 2014). Academic studies further document that such deceptive feedback distorts consumer beliefs, inflates product ratings, and undermines market efficiency.

In parallel, research on spam review detection has evolved from rule-based and feature-engineered approaches toward more advanced deep learning models, including convolutional and recurrent neural networks as well as transformer-based language models. Recent studies show that neural architectures can effectively capture contextual semantics, stylistic irregularities, and intent-level cues associated with deceptive reviews, often outperforming traditional classifiers (Adelani et al., 2020; Hou et al., 2025; Salminen et al., 2022). These advances underscore the growing importance of representation learning in identifying deceptive review content and directly motivate RQ1, which focuses on how spam reviews can be effectively identified and operationalized as credibility signals.

To illustrate common patterns of deceptive reviews and their potential impact on platform-level information processing, we present two representative examples drawn from the labeled dataset used in this study (Saxena, 2018). These examples serve as theory-informed vignettes that highlight distinct linguistic and behavioral characteristics of opinion spam.

Example A: Vacuous promotional praise (Label: Spam)

Title: Best purchase ever!!!!!

Rating: ★★★★★

Content: "This is the best product I have ever bought in my life. Highly recommended to everyone. You must buy it now—don't hesitate. Great quality and fast shipping."

Despite its strongly positive sentiment, this review provides no verifiable product-specific information (e.g., functionality, durability, or comparative usage experience). Prior research characterizes such content as promotional opinion spam, often associated with incentivized or coordinated campaigns rather than genuine consumption experiences (Mayzlin et al., 2014; Z. Wu et al., 2024).

Example B: Competitive defamation (Label: Spam)

Title: Don't waste your money.

Rating: ★☆☆☆☆

Content: "Worst experience ever. The item arrived broken and customer service was terrible. Save your money and buy [Competitor Brand] instead—it's much better than this trash."

This review exemplifies strategic negative spam, in which exaggerated complaints are combined with explicit redirection toward a competitor. Such reviews have been documented as a mechanism for competitive manipulation rather than authentic post-purchase evaluation (S. Choi et al., 2017; He et al., 2022).

Importantly, these examples also illustrate how deceptive signals—such as exaggerated sentiment, generic praise, or strategically biased content—may be treated as equally credible inputs in recommender systems. As a result, such signals can distort product evaluation and personalized ranking outcomes if not properly accounted for. This observation motivates not only the need for effective spam detection but also the integration of credibility-aware mechanisms into downstream recommendation processes.

These vignettes reflect recurring deceptive strategies discussed in prior literature and motivate the linguistic and behavioral features identified in spam detection research. More importantly, spam reviews do not merely mislead individual consumers; when treated as equally credible inputs, they can systematically bias rating aggregation and review-based recommendation pipelines.

At the same time, recommender systems play a pivotal role in digital marketplaces by transforming review information into personalized product rankings. Through collaborative filtering, content-based modeling, and hybrid approaches, recommender systems shape product visibility and user exposure. However, most recommender systems implicitly assume that all reviews are equally credible sources of information. When spam reviews contaminate the input data, algorithmic personalization may inadvertently amplify distorted signals.

Existing academic research has largely evolved along two separate streams. One stream focuses on spam review detection, leveraging linguistic, sentiment, and behavioral features to classify deceptive content (Hussain et al., 2019; Mukherjee et al., 2013; Paul & Nikolaev, 2021). The other stream focuses on recommendation systems that optimize personalization based on user–item interactions (Koren et al., 2009). While prior studies have begun to incorporate

credibility or trust signals into recommender systems, such as trustworthy and robust recommendation models that address noisy or malicious inputs (e.g., Wang et al., 2024a, 2024b), as well as recent efforts that jointly integrate fake review detection with recommendation (e.g., Guo et al., 2025), these approaches typically rely on tightly coupled or model-specific architectures. For example, credibility is often embedded within collaborative filtering models or incorporated through joint optimization frameworks, rather than treated as a transferable signal across stages.

Consequently, few studies have operationalized credibility as a continuous signal that can be explicitly propagated from the detection stage into downstream recommendation processes, leaving a critical gap between spam detection and recommendation outcomes. To address this gap, this study conceptualizes review credibility as an actionable signal that is propagated into downstream recommendation processes, rather than treated as an isolated detection outcome.

Building on this perspective, we propose a spam-aware, credibility-informed recommendation framework. Specifically, we first identify spam reviews using machine learning and deep learning models incorporating semantic, sentiment, and metadata features. We then integrate reviewer credibility signals into a hybrid recommendation model to improve recommendation accuracy and robustness. Accordingly, the goal of this study is not merely to classify spam reviews, but to enhance personalized recommendations by grounding them in credible review information.

This study addresses the following research questions:

- RQ1: How can spam reviews be identified and operationalized using linguistic, sentiment, and behavioral features?
- RQ2: How does excluding or down-weighting spam reviews affect the accuracy and reliability of personalized recommendations?
- RQ3: How can reviewer credibility signals be incorporated into recommendation models to enhance user trust and decision quality?

The remainder of this paper is organized as follows. The "Related literature" section reviews related literature on online review credibility, spam detection, and recommendation systems. The "Research framework and methodologies" section introduces the proposed integrated framework. The "Experiments" section explains the data preparation and experimental setup. The "Results and evaluation" section reports evaluation results. The "Discussion and conclusion" section concludes with implications and future research directions.

Related literature

This section reviews prior research on online review credibility and spam review detection. Building on comprehensive syntheses of the field (e.g., Wu et al., 2020), we discuss review credibility under information overload, recent advances in spam detection, and implications for recommender systems, laying the foundation for the proposed credibility-weighted recommendation framework.

Review credibility and the challenge of information overload

As online platforms increasingly rely on user-generated content, understanding how consumers evaluate review credibility under conditions of information abundance has become a central research concern.

Importance and impact of online reviews

The rapid proliferation of user-generated content has transformed traditional word-of-mouth (WOM) into electronic word-of-mouth (e-WOM), enabling consumers to share experiences and opinions in real-time (Chan & Ngai, 2011; Litvin et al., 2008). Online consumer reviews (OCRs) have become a vital resource for evaluating products and services (H. S. Choi & Leon, 2020; Daugherty & Hoffman, 2014). However, the sheer volume and varying quality of these reviews frequently lead to information overload, which can hinder consumers' ability to extract meaningful insights and make informed decisions. As Hu and Krishen (2019) contend, when the quantity of available reviews exceeds a certain threshold, consumers may experience cognitive fatigue, reducing the overall helpfulness of the feedback.

Prior research suggests that excessive volumes of redundant or low-quality reviews can erode consumer trust and reduce decision confidence. For example, Furner & Zinko (2017) show that review overload diminishes purchase intention by lowering perceived trustworthiness, while Hu and Krishen (2019) argue that cognitive fatigue reduces consumers' ability to extract diagnostic information. Multimedia-enhanced reviews may further intensify overload when not effectively curated (L. Wang et al., 2024a, 2024b).

Platform-level review curation mechanisms and their limitations

To mitigate review overload and help consumers surface diagnostic information, platforms deploy a set of user-facing curation mechanisms. Common examples include

helpful-vote rankings, star-rating summaries, verified purchase badges, reviewer profile cues, as well as sorting and facet-based filtering options that allow users to prioritize reviews by criteria such as quality, durability, or functionality (Askalidis et al., 2017; Ganguly et al., 2024; Mukherjee et al., 2013). More recently, platforms have introduced AI-generated review summaries to reduce reading costs by condensing large volumes of feedback into concise takeaways (L. Wang et al., 2024a, 2024b). Collectively, these mechanisms shape both visibility (which reviews are surfaced) and interpretation (which reviews appear most diagnostic), thereby functioning as practical governance tools in large-scale review environments.

Among these mechanisms, helpful votes have been extensively studied as indicators of perceived review quality or usefulness. Prior research shows that helpfulness votes capture how informative, well-articulated, or persuasive a review appears to other users, and they play a meaningful role in directing consumer attention and reducing search costs in review-dense environments (Siering et al., 2018; Yu et al., 2022). From a user-interface perspective, such signals can therefore improve navigability and support decision-making.

However, prior research consistently demonstrates that helpful votes are not equivalent to ground-truth credibility signals and may be subject to systematic bias. In particular, helpful-vote rankings and "most helpful" displays exhibit exposure-related dynamics, whereby early visibility attracts additional votes and creates persistence advantages for already prominent reviews (Askalidis et al., 2017; P. Liu et al., 2017). Empirical studies further document social influence and herding effects in review evaluation, in which prior votes shape subsequent judgments independent of intrinsic review quality (Chevalier & Mayzlin, 2006; J. Lee et al., 2008; Zacharia et al., 2000). As a result, well-written but deceptive reviews may accumulate high helpfulness scores, while authentic yet less visible reviews remain underweighted.

However, helpful votes are not equivalent to ground-truth credibility signals. Prior studies show that "most helpful" rankings are shaped by exposure effects, popularity reinforcement, and herding dynamics, which may amplify persuasive but deceptive reviews (Askalidis et al., 2017; Chevalier & Mayzlin, 2006; P. Liu et al., 2017). Moreover, helpfulness evaluations vary across product categories and decision contexts, limiting their reliability as general-purpose credibility proxies (Siering et al., 2018; Yu et al., 2022). Similarly, verified purchase badges, star histograms, and AI-generated summaries can reduce reading costs and improve navigability, but they remain interface-layer cues and do not generate machine-consumable credibility signals for recommendation learning (Mukherjee et al., 2013; L. Wang et al., 2024a, 2024b). Consequently, platform curation mechanisms alleviate overload primarily at the presentation layer, without resolving the structural problem of ensuring that personalization models learn from credible feedback.

Spam review detection

Online review platforms have become essential channels for sharing product and service evaluations. However, their open nature has also fostered the proliferation of fake or misleading reviews—often termed spam reviews—which are deliberately posted to manipulate consumer perceptions (Askalidis et al., 2017). Early foundational work by Jindal and Liu (2007, 2008) categorized spam reviews into untruthful opinions (e.g., overly positive or negative reviews), brand-centric promotion, and irrelevant or advertisement-like content. Subsequent surveys refined these typologies and highlighted the intrinsic difficulty of deception detection, documenting that even human evaluators achieve limited accuracy when distinguishing nuanced spam from genuine feedback (Hussain et al., 2019; Mohawesh et al., 2021; Ott et al., 2013).

Wu et al. (2020) synthesize behavioral, economic, and computational perspectives on fake online reviews and identify three dominant streams: (1) motivations and incentives underlying deceptive reviewing, (2) feature-based and machine-learning detection methods, and (3) platform-level governance mechanisms. Importantly, they note that most studies treat detection accuracy as the primary endpoint, with limited attention to how detection outputs affect downstream ranking, recommendation, or personalization mechanisms.

Early detection approaches relied primarily on feature-engineered, content-based methods, leveraging linguistic patterns, sentiment polarity, and semantic similarity to identify anomalous reviews (Fei et al., 2013; Li et al., 2011). Complementary reviewer-based approaches examined behavioral signals such as posting frequency, burstiness, and temporal coordination to identify suspicious accounts or campaigns (Cresci et al., 2015). While these methods demonstrated practical effectiveness, they often treat textual and behavioral signals in isolation, leaving coordinated spam strategies able to exploit single-modality weaknesses.

More recent research reflects a shift toward deep learning and representation learning frameworks. Studies adopting convolutional, recurrent, and transformer-based architectures demonstrate improved ability to capture contextual semantics and stylistic irregularities associated with deceptive content (Deshai & Rao, 2022; Hajek et al., 2023). Importantly, Adelani et al. (2020) show that neural language models can generate sentiment-preserving fake reviews that closely resemble authentic content, substantially increasing the difficulty of detection and challenging traditional lexical heuristics. Salminen et al. (2022) further conceptualize fake

review generation and detection as a co-evolving adversarial process, emphasizing that increasingly fluent generators necessitate more expressive and robust detection strategies.

Recent work also extends spam detection beyond purely textual analysis. Hou et al. (2025) propose a multimodal deep learning framework that integrates textual, contextual, and behavioral cues to infer deceptive review intentions, while Alghaligah et al. (2025) develop hybrid deep architectures that combine representation learning with optimization techniques to enhance robustness across product categories. Collectively, these advances signal a transition from surface-level linguistic cues toward intent-aware and representation-driven detection frameworks.

In line with this literature, prior empirical studies consistently identify recurrent deception cues across spam strategies. These include sentiment extremeness (Ott et al., 2013), review conciseness and topical redundancy (Fei et al., 2013), semantic similarity across review sets (Li et al., 2011), and metadata authenticity indicators such as verified purchase flags (Mukherjee et al., 2013). More recently, research on large language model–generated reviews highlights the relevance of readability and stylistic regularity metrics as additional discriminative signals (Pandey & Singh, 2025). Table 1 summarizes the major feature categories adopted in this study, representative operational signals, and their grounding in prior research. To ensure conceptual clarity, we distinguish between fine-grained feature categories derived from prior literature and higher-level feature families used in the system design. Specifically, the six categories identified in this section—textual/linguistic features, sentiment, semantic similarity, readability/style, metadata authenticity, and reviewer behavior—represent a literature-driven taxonomy.

For implementation purposes, these categories are systematically grouped into four broader feature families in the "Research framework and methodologies" section. In this mapping, textual/linguistic features, semantic similarity, and readability/style are subsumed under the content family, while sentiment, metadata, and reviewer behavior are retained as distinct families. This hierarchical structure preserves the richness of prior studies while enabling a more parsimonious and modular system design.

Table 1 presents a hierarchical feature taxonomy that distinguishes between fine-grained categories derived from prior literature and higher-level feature families used in our framework. This structure ensures conceptual alignment between the literature review and the system design described in the "Research framework and methodologies" section.

Despite substantial progress, many existing detection approaches remain task-isolated, concluding at classification accuracy without examining how detection outputs affect downstream personalization or recommendation outcomes. This limitation is particularly salient in large-scale platforms, where even small detection errors can propagate through recommendation pipelines. Building on the above insights, our study adopts a hybrid detection design that integrates content (including textual, semantic, and readability features), sentiment, metadata, and reviewer-behavior signals, explicitly motivated by the need to support credibility-aware recommendation.

An important emerging frontier concerns AI-generated opinion spam. Beyond human-authored deception, recent studies demonstrate that large language models can synthesize highly fluent, sentiment-consistent reviews that bypass traditional lexical heuristics and blur the boundary between authentic and deceptive content (Adelani et al., 2020; Salminen et al., 2022). This development highlights a fundamental limitation of text-only detection strategies in isolation.

Although the present study does not explicitly train a dedicated AI-generated spam detector, our framework remains robust under AI-assisted spam scenarios by emphasizing behavioral signals (e.g., posting burstiness, cross-item

Table 1 Hierarchical feature taxonomy, representative signals, and prior foundations

Feature family	Fine-grained category	Representative signals	Supporting literature	Role in this study
Content	Textual/linguistic	Review length, lexical diversity	Jindal and Liu (2008); Ott et al. (2013)	Baseline content cues
	Semantic similarity	TF–IDF similarity, cosine similarity	Li et al. (2011); Fei et al. (2013)	Detects repetitive or synthetic content
	Readability/style	Readability scores, stylistic regularity	Pandey and Singh (2025)	Identifies LLM-like patterns
Sentiment	Sentiment based	Polarity extremeness, subjectivity	Hajek et al. (2023)	Detects exaggerated persuasion
Metadata	Metadata authenticity	Verified purchase, rating plausibility	Mukherjee et al. (2013)	Captures buyer authenticity signals
Reviewer Behavior	Reviewer behavior	Burstiness, cross-item similarity	Cresci et al. (2015); Hussain et al. (2020)	Scalable credibility modeling

similarity, and temporal coordination) that are orthogonal to textual fluency and remain informative across both human- and AI-generated manipulation.

Recommendation systems

Recommendation systems have become a pivotal tool for alleviating information overload by providing users with personalized suggestions drawn from vast data sources (Q. Liu et al., 2024; Mao et al., 2016; Migliorini et al., 2024). Personalized recommendations began in the 1990 s and are now crucial for e-commerce platforms. Major approaches include content-based filtering, which compares user interests with item attributes using techniques such as vector space models or k-nearest neighbors (Pazzani & Billsus, 2007), and collaborative filtering, which infers user preferences by analyzing behavioral similarities among users or items (Ekstrand et al., 2011; H.-N. Kim et al., 2010). Query suggestion refines user queries by examining search patterns (Varma et al., 2015), while hybrid methods combine multiple techniques to address the shortcomings of any single approach, often achieving higher accuracy (Yun et al., 2018).

Despite their success in e-commerce, existing recommendation systems rarely account for the reliability of the underlying reviews that inform their algorithms. Many assume that aggregated user ratings or textual reviews accurately reflect product quality, overlooking the potential contamination of fake or biased feedback (Varma et al., 2015). This oversight not only risks misleading users but can also erode trust in recommendation platforms. Recent studies highlight that spam reviews can systematically skew item ratings, rendering naive recommendation models vulnerable to manipulation (Y. Liu, 2020; Zhang et al., 2023).

Furthermore, emerging research suggests that consumers exhibit growing skepticism toward online reviews (Kollmer et al., 2022). When users suspect that reviews are manipulated, they may lose confidence in both the product and the platform, reducing engagement with recommendation systems. Wang et al., (2024a, 2024b) propose that trustworthy recommender systems should incorporate mechanisms to assess review credibility, but existing implementations remain limited. Similarly, Xia et al. (2021) explore anomaly-based filtering techniques for improving recommendation quality, though their focus is primarily on identifying outliers rather than detecting fraudulent content.

In addition to accuracy and credibility, fairness is another critical challenge. Jin et al. (2023) highlight that biased or manipulated reviews can disproportionately influence recommendation rankings, leading to unfair exposure for certain products or sellers. However, current fairness-aware recommender models rarely integrate explicit spam detection mechanisms.

Given these gaps, it is essential to develop a recommendation framework that filters spam reviews and ensures that recommendations are derived from credible, high-quality user feedback. This study addresses, this issue by integrating spam detection with a hybrid recommendation approach, thereby enhancing both the reliability and fairness of e-commerce recommendations.

Toward spam-aware recommendation

Taken together, prior research underscores the importance of trustworthy online reviews and highlights the risks posed by spam contamination. Although both spam detection and recommendation systems have been extensively studied, these two streams have largely evolved in parallel. Many recommender systems implicitly assume that all reviews are equally reliable (S. Wang et al., 2024a, 2024b) or rely on heuristic user-interface mechanisms—such as helpfulness voting, star histograms, or popularity-based filtering—to manage review quality (Askalidis et al., 2017). While anomaly-based models (Xia et al., 2021) and fairness-aware recommenders (Jin et al., 2023) improve ranking robustness, they do not explicitly model the structural distortion introduced by deceptive review content.

Recent work has begun to bridge spam detection and recommendation through trust-aware and spam-aware recommender systems. For example, Dong et al. (2022) survey deep learning–based trust-aware recommender models that incorporate user trust, social relationships, or reliability signals into recommendation learning. Similarly, Guo et al. (2025) propose a mixture-of-experts framework that jointly optimizes fake review detection and recommendation tasks. These approaches establish the importance of treating ratings and reviews as non-uniformly reliable signals.

However, existing approaches typically adopt a model-centric coupling strategy, in which detection and recommendation are tightly integrated through joint optimization or task-specific architectures. While effective, such designs may limit deployability, interpretability, and modular reuse, particularly in large-scale platforms where recommender pipelines are already operational.

To provide a clearer synthesis of existing mechanisms, Table 2 summarizes representative research streams addressing review overload, credibility assessment, and recommendation integration. The table highlights that most prior approaches operate either at the user-interface layer or treat spam detection as an isolated back-end task, whereas relatively few studies operationalize credibility as a reusable input to personalization.

Against this background, we propose a spam-aware, credibility-weighted recommendation framework that adopts a

Table 2 Comparison of review curation, spam detection, and credibility-aware recommendation (RS denotes recommender system)

Approach category	Typical layer	Representative studies/practice	Role in managing reviews	Key limitation	Position of this study
UI-level review curation mechanisms (helpful votes, stars, badges, filters, summaries)	UI/presentation	Chevalier and Mayzlin (2006); Askalidis et al. (2017); Mukherjee et al. (2013); Ganguly et al. (2024); L. Wang et al., (2024a, 2024b)	Reduces cognitive load and improves navigability	Reflect perceived usefulness or popularity, not verifiable credibility; prone to exposure and herding bias	Treated as complementary cues, not credibility weights
Spam review detection (content/behavior)	Back-end	Jindal and Liu (2008); Ott et al. (2013); Wu et al. (2024); Salminen et al. (2022); Hou et al. (2025)	Identifies deceptive or manipulative reviews	Typically isolated from recommendation learning	Detection outputs are operationalized for personalization
Trust-aware recommender systems	Back-end + RS	Dong et al. (2022)	Models trust, reliability, or user relationships	Trust signals often indirect or user-centric	Complements by focusing on review-level credibility
Joint detection–recommendation models	Back-end + RS (joint)	Guo et al. (2025)	End-to-end optimization of detection and recommendation	Tightly coupled architectures reduce modularity and deployability	Our approach provides a modular alternative
Credibility-weighted recommender (this study)	Back-end + RS	—	Injects review-level credibility weights into ranking	Requires labeled detection training	Bridges spam detection and personalization via auditable weights

complementary design philosophy to existing approaches. Rather than tightly coupling spam detection and recommendation through joint end-to-end optimization, our framework explicitly separates detection from personalization and operationalizes review- and reviewer-level credibility assessments as continuous, auditable weights. These weights are subsequently injected into hybrid recommendation pipelines to reshape personalized ranking.

This design choice addresses key limitations identified in prior work. While joint detection–recommendation models demonstrate promising performance, they often require tightly integrated architectures and task-specific retraining, limiting modularity and deployability. In contrast, our approach preserves compatibility with existing recommender infrastructures while allowing credibility signals to systematically influence recommendation outcomes.

As summarized in Table 2, our framework bridges the gap between review authenticity and personalization by moving beyond user-interface heuristics and isolated spam classification. By treating credibility as a machine-consumable signal rather than a binary filter or heuristic proxy, the proposed approach enables more robust Top-N recommendations under realistic spam contamination, without necessitating architectural redesign. This positioning distinguishes our contribution from both interface-level curation mechanisms and prior trust-aware or joint learning models.

Research framework and methodologies

This study proposes a spam-aware hybrid recommendation framework that integrates opinion mining, sentiment analysis, and spam detection to reduce the influence of deceptive reviews in e-commerce environments. The framework is modular and theory-informed, linking established deception cues with downstream recommendation mechanisms.

Building on prior research in spam detection and recommender systems, the proposed architecture integrates four feature families—content, sentiment, metadata, and reviewer behavior. These correspond to deception cues widely supported in the literature, including linguistic redundancy (Ott et al., 2013), sentiment extremeness (Hajek et al., 2023), metadata authenticity (Mukherjee et al., 2013), and behavioral irregularity (Cresci et al., 2015). This design relaxes the common assumption in recommender systems that all review signals are uniformly reliable.

To ensure consistency between the Background (Table 1) and the implemented framework, we adopt a hierarchical feature taxonomy. Table 1 summarizes fine-grained

categories commonly used in spam detection (textual/linguistic cues, sentiment, semantic similarity, readability/style, metadata authenticity, and reviewer behavior). For implementation clarity, these categories are consolidated into four higher-level feature families: (i) content (textual cues, semantic similarity, readability/style), (ii) sentiment, (iii) metadata, and (iv) reviewer behavior. This mapping preserves the breadth of prior research while enabling a parsimonious and modular system design.

The overall process, illustrated in Fig. 1, follows four major stages:

1. Review collection and spam detection: We collect product reviews and apply feature engineering to identify spam reviews.
2. Reviewer behavior analysis: We analyze reviewer behavior to estimate reviewer-level suspiciousness and down-weight reviews from low-credibility users (and, in a robustness check, optionally remove them) to enhance overall review quality.
3. User preference analysis: We analyze user reviews and ratings using opinion mining and construct a user preference matrix to infer personalized preferences.
4. Personalized recommendation generation: We combine the user preferences with quality reviews to produce a tailored product recommendation list for each user.

Figure 1 illustrates the end-to-end workflow, showing how review credibility assessments are propagated into downstream personalization. A key contribution of this study is credibility-weighted personalization, which operationalizes spam detection outputs as continuous signals that directly affect recommendation learning. Rather than treating spam detection as a terminal classification task, credibility is propagated into product scoring and personalized ranking.

Formally, for each review r_i written by reviewer u, the review-filtering model outputs a spam likelihood $p_i = P(\text{spam} \mid r_i)$, which is transformed into a review credibility score $c_i^{rev} = 1 - p_i$. In parallel, the reviewer-detection model estimates a spammer likelihood $q_u = P(\text{spammer} \mid u)$, yielding a reviewer credibility score $c_u^{usr} = 1 - q_u$. These two signals are combined into a single credibility weight for each review: $w_{iu} = c_i^{rev} \times c_u^{usr}$, where $w_{iu} \in [0,1]$.

This weight is subsequently used to (i) compute credibility-weighted product trust and aspect scores, and (ii) weight review-derived preference signals in the hybrid recommendation stage. Through this mechanism, suspicious reviews and suspicious reviewers exert proportionally less influence on personalization outcomes.

The complete system consists of five modules (Fig. 2), ranging from data preparation to credibility-weighted recommendation generation.

1. Data preparation module: This module handles data collection, sampling, and extraction from Amazon product and customer review datasets. It provides the input for subsequent analysis and recommendation.
2. Review filtering module: This module distinguishes spam reviews from genuine ones by examining content features (e.g., writing style, use of emotional language), sentiment, and metadata.
3. Reviewer detection module: This module identifies potential spammers by analyzing review repetition and temporal patterns in posting behavior.

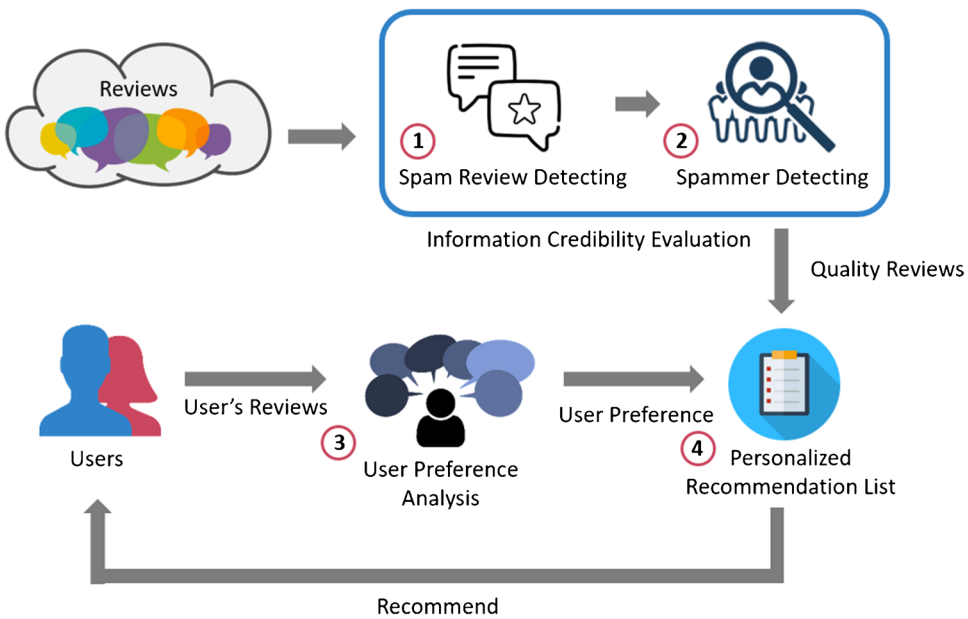


Fig. 1 The process of our mechanism

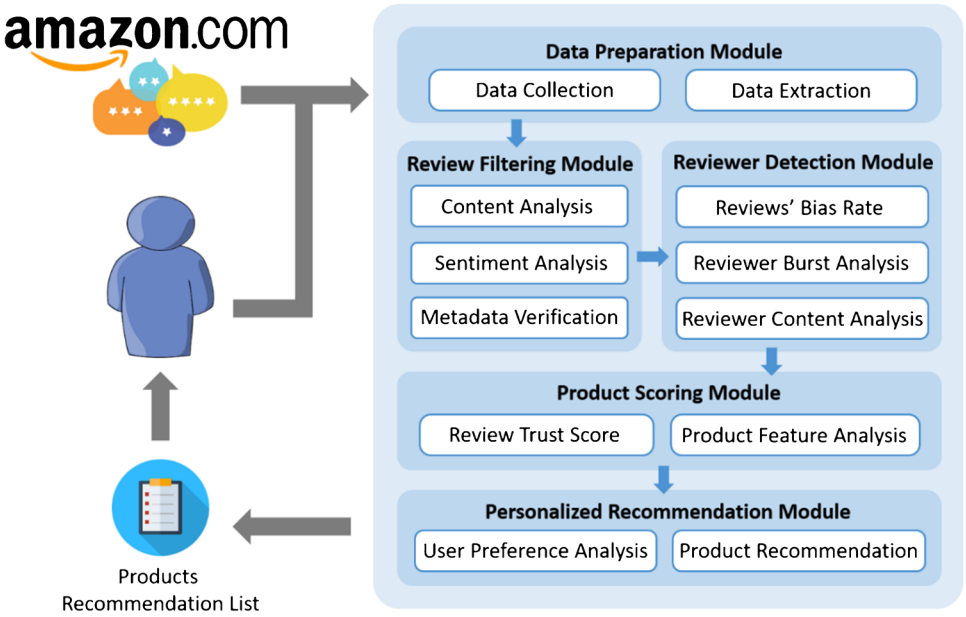


Fig. 2 The system framework

4. Product scoring analysis module: In this module, product ratings are recalculated after filtering out spam and spammer reviews. Sentiment analysis is then used to compute aspect scores (e.g., quality, cost, service, brand) for personalized recommendations.
5. Personalized recommendation module: This module aggregates user preferences and high-quality review data from the previous modules to generate a final, personalized recommendation list for each user.

The following sections detail each module in greater depth.

Data preparation module

As the foundation of the system, this module handles data collection, sampling, and extraction from Amazon product and customer review datasets. Reviews, ratings, timestamps, and reviewer IDs are standardized through tokenization, stop-word removal, stemming, and punctuation filtering using Python's NLTK and TextBlob libraries. This preprocessing ensures a clean and consistent dataset for subsequent modules and supports cross-module comparability (Mukherjee et al., 2013; Paul & Nikolaev, 2021).

Data collection

We employ two types of data in our research—one set for training the spam review detection model and another for developing the recommendation system. Most Amazon product review datasets are either unlabeled or labeled based on ratings or helpfulness. Previous studies have generated labeled datasets by manually annotating reviews or by hiring professional writers to produce both genuine and fake reviews. Although many researchers use labeled Yelp hotel reviews for training (Shehnepoor et al., 2017), we focus on product reviews to better align with our application. Given the high volume of daily reviews, manual labeling is impractical. Fortunately, a recently published dataset of labeled Amazon reviews for deception detection (Saxena, 2018) is available and is used to train our review filtering module. Additionally, we collect detailed unlabeled product reviews(Jindal & Liu, 2008; Ni et al., 2019) which are primarily used in the reviewer detection module.

The labeled dataset (Saxena, 2018) was retrieved from a publicly available GitHub repository. The unlabeled Amazon review corpora are the publicly available datasets described by Jindal and Liu (2008) and Ni et al. (2019). All datasets are publicly available for research and reproducibility purposes.

Data extraction

Data extraction involves two primary objectives: reducing the dataset size and filtering content based on relevance. First, due to the enormous volume of the unlabeled dataset, we extract only the most recent reviews to enhance experimental efficiency, as older data is less useful for our recommendation system. In contrast, the dataset containing 21,000 labeled reviews is used in its entirety. Second, we clean the data by removing null entries and non-English reviews, and we select only the necessary columns for our experiments. This step minimizes data loading and improves overall processing efficiency.

Review filtering module

Building on the prepared dataset, this module identifies deceptive reviews using a multi-source feature representation that integrates content, sentiment, metadata, and style-based signals:

- Content features: textual/linguistic cues (e.g., pronoun ratios, review length, n-grams), semantic similarity signals (e.g., TF–IDF/Word2Vec cosine similarity) (Fei et al., 2013; Li et al., 2011), and readability/style metrics (e.g., readability scores and stylistic regularity);
- Sentiment features: polarity and subjectivity scores capturing exaggerated emotional tone (Hajek et al., 2023);
- Metadata features: verified-purchase indicators and rating–sentiment mismatches serving as authenticity cues (Mukherjee et al., 2013).

These feature categories are combined to estimate a spam likelihood p_i for each review, which is transformed into a review credibility score c_i^{rev}. This score is later combined with reviewer-level credibility to form the final credibility weight used in product scoring and recommendation.

Content analysis

Content feature analysis focuses on the writing style of users. Prior studies, such as those by Li et al. (2011), indicate that spam reviews contain fewer first-person terms compared to genuine ones, while featuring more second-person and third-person terms. Consequently, Li et al. utilized the ratio of first-person to second-person pronouns as a distinguishing feature. Akram et al. (2018) expanded on this by including third-person pronouns, calculating the ratio of the sum of second- and third-person pronouns to first-person pronouns. In our analysis, we will track the counts of first-person (FP), second-person (SP), and third-person (TP) pronouns, as well as the following ratios:

FSP: ratio of second-person to first-person pronouns $FSP = \frac{N_{SP}}{N_{FP}}$

FSTP: ratio of the sum of second- and third-person to first-person pronouns

$$FSTP = \frac{N_{SP} + N_{TP}}{N_{FP}}$$

Next, we will assess the length of comments. Research by Crawford et al. (2015) suggests that spam reviews tend to be shorter, as spammers lack genuine experiences and do not invest time in writing detailed feedback. Therefore, we compute the length of each review

$$RL_i = len(review_i)$$

Additionally, spammers often copy existing reviews. Mukherjee (2013) examined content similarity, while Fei (2013) employed an average similarity formula. We will calculate the cosine similarity between reviews to detect similarities:

$$RCS(r_i, r_j) = cosine(r_i, r_j)$$

The composite score, csScore, is derived from the top 10 most similar reviews:

$$csScore = \frac{\sum_{i=1}^{10} RCS_{topi} * label_{topi}}{\sum_{i=1}^{10} RCS_{topi}}$$

Here, RCS represents the cosine similarity of other reviews, with a true review labeled as 1 and a fake review as − 1. A score closer to 1 indicates a higher likelihood of authenticity, while a score closer to − 1 suggests spam.

We utilize Word2Vec for cosine similarity calculations. Previous studies have demonstrated the effectiveness of n-gram-based approaches in detecting spam reviews, so we will incorporate both Word2Vec and TF-IDF for further feature engineering. TF-IDF (term frequency–inverse document frequency) measures how important a word is within a collection of reviews based on how frequently it appears, whereas Word2Vec creates word embeddings—numerical vector representations of words that capture semantic relationships, allowing the model to understand contextual similarity between terms.

Unigram Count: $Uni(review_i) = \#((wordsinreview_i) \in Dict(Unigram))$

Bigram Count: $Bi(review_i) = \#((wordsinreview_i) \in Dict(Bigram))$

Trigram Count: $Tri(review_i) = \#((wordsinreview_i) \in Dict(Trigram))$

Next, we calculate the TF-IDF score for each word in the reviews:

$$\text{score}_{t,i} = tf_{t,i} \times idf_t$$

where t denotes a particular term t and i denotes a particular review i.

$$tf(t,i) = \frac{f_{t,i}}{\sum_{t' \in i} f_{t',i}}$$

where $tf_{t,i}$ denotes the term frequency adjusted for review length and $f_{t,i}$ denotes the number of times term t appears in review i.

$$idf(t,i) = \log \frac{N}{|\{i \in I : t \in i\}|}$$

where $idf(t,i)_i$ denotes the inverse document frequency of term t, N denotes total number of reviews in the corpus, and $|\{i \in I : t \in i\}|$ denotes number of reviews where the term t appears ($tf(t,i) \neq 0$).

Sentiment analysis

We employ sentiment analysis to evaluate the users' comments. Previous research (Li et al., 2011; Mekouar et al., 2014) indicates that spam reviews often exhibit more extreme emotions, either excessively positive or negative. We will utilize the TextBlob toolkit in Python to compute the polarity score (PC) and subjectivity score (SC):

Polarity: $PC_i = Polarity(review_i)$

Subjectivity: $SC_i = Subjectivity(review_i)$

In addition to detection, sentiment cues are retained because exaggerated persuasion patterns can propagate into product scoring and ranking if left unfiltered.

Metadata verification

Research by Li (2011) and Mukherjee (2013) highlights the importance of metadata in enhancing the accuracy of spam review detection. In this study, we utilize metadata indicating whether a reviewer is a verified purchaser, which serves as a proxy for transactional authenticity and provides an additional signal beyond text-based cues.

$$VP_i = VerifiedPurchase(review_i)$$

Metadata cues are particularly relevant here because verified purchases provide a transactional proxy that is difficult to falsify at scale, offering a cross-check beyond text-only heuristics.

Reviewer detection module

Building on review-level detection results, this module estimates reviewer suspiciousness using behavioral indicators such as bias rate, posting burstiness, and cross-product repetition (Hussain et al., 2020; P. Liu et al., 2017). The module outputs a spammer likelihood q_u, which is converted into reviewer credibility $c_u^{usr} = 1 - q_u$ and combined with review credibility to form the final weight w_{iu}.

Reviews' bias rate

In their research, Liu et al. (2017) identified bias rate as a crucial indicator of spam activity. This metric counts the number of reviews left by the same user on the same product, referred to as the bias rate. They suggest that a second review may reflect a different experience than the first, making it potentially more trustworthy. However, if a reviewer posts three or more reviews for the same product, they may be identified as a spammer. In our analysis, we observed that bias rates not equal to one are often posted on the same day. Therefore, we calculate the number of posts by a specific reviewer on a product within a given day, adding a temporal condition for the second review.

$$UPT_{p,u,t} = posttimes_{p,u,t}$$

where p denotes a product, u denotes a user, and t denotes time. Since multiple reviews may indicate bias, some e-commerce platforms restrict users to two reviews per product. Our dataset does not impose this restriction, so we implement it for our analysis.

Review burst analysis

Previous studies (Fei et al., 2013; Hussain et al., 2020; Mukherjee et al., 2013) have shown that spam reviews can also be identified through reviewer behavior, particularly review burstiness. A burst review occurs when a reviewer submits numerous reviews for different products within a short timeframe. Spammers often generate a large volume of comments (P. Liu et al., 2017; Mukherjee et al., 2013) in a short period to influence customer opinions or manipulate perceptions. Therefore, we classify a reviewer who submits many reviews in a brief time as abnormal. We analyze the maximum number of comments, submitted by a reviewer within a specific time frame to detect spam activity.

$$UBR_{u,t} = max(\#review_u int)$$

where u denotes a user, and we assess the maximum number of reviews, submitted by that user during the time period t.

Reviewer content analysis

We analyze reviewer content to determine if a reviewer is a spammer. Review length is a key indicator of spam reviews. In studies by Hussain et al. (2020) and Liu et al. (2017), it was found that English reviews with fewer than five words are often classified as spam. We observe that if a reviewer consistently leaves short reviews, they may be a spammer or, at the very least, their reviews may lack substance.

$$UL_{u,i} = len(review_i)$$

where u denotes a user and i denotes the review i of the user u.

In addition to review length, we calculate the cosine similarity among a reviewer's own reviews. As noted in spam detection studies (Gera et al., 2015), spammers often exhibit similar phrasing across multiple reviews. Specifically, a spammer may leave identical reviews for the same product or similar reviews across different products. Our research calculates the similarity of all reviews from a user. High cosine similarity among a reviewer's submissions suggests

spam behaviour, indicating that these reviews are not based on genuine product experiences.

$$UCS_u = \frac{\sum cosine\left(r_{u1}, r_{u2}, \ldots, r_{un}\right) - 1}{numberofreviewsfromu}$$

where u denotes a user, and we compute a cosine similarity metric for all reviews r_{u1} to r_{un}, submitted by that user, yielding a cosine similarity score.

Product scoring analysis module

Using credibility weights from the prior modules, this stage computes a credibility-weighted Review Trust Score (RTS) for each product and derives aspect-level sentiment scores. By reducing noise propagation prior to recommendation generation, this module improves the reliability and interpretability of downstream ranking results (Hajek et al., 2023).

Review trust score

After selecting the important features, we employ several machine learning models, including logistic regression and random forest, as well as deep learning models like LSTM, to train our labeled data and identify the best-performing model for the next step. We assess the effectiveness of these machine learning methods using statistical metrics such as precision, recall, and F1-score. The confusion matrix, a standard tool for evaluating classification models, is illustrated in Table 3.

Following prior spammer-detection research (Hussain et al., 2020), we include a composite behavioral indicator (SpammerScore) as an interpretable reviewer-level feature summarizing posting bias (UPT), burstiness (UBR), review length abnormality (UL), and intra-user similarity (UCS). Importantly, reviewer suspiciousness is determined by the supervised model output $q_u = P(\varepsilon spammer \varepsilon | u)$ rather than by applying a deterministic threshold on SpammerScore.

In this matrix, "1" (positive) indicates a non-spam review, while "0" (negative) indicates a spam review.

Precision measures the accuracy of positive predictions: $\text{Precision} = \frac{TP}{TP+FP}$

Recall reflects the ability to identify actual positives: $\text{Recall} = \frac{TP}{TP+FN}$

Table 3 Confusion matrix

	Predict: 1 (positive)	**Predict: 0 (negative)**
Actual: 1 (positive)	True positive (TP)	False positive (FP)
Actual: 0 (negative)	False negative (FN)	True negative (TN)

F1-Score combines precision and recall into a single metric: $\text{F1} - \text{Score} = \frac{2}{\frac{1}{\text{Precision}} + \frac{1}{\text{Recall}}}$

$$\text{SpammerScore} = UPT + UBR + UL + UCS$$

where UPT refers to the user post-time for a specific product. UL refers to the user review length. UCS refers to the cosine similarity of a user's reviews and UBR refers to the user's review burstiness.

Product feature analysis

Opinion mining, or sentiment analysis, is employed to identify positive and negative sentiments from many reviews. Our goal in this analysis is to extract key product features and calculate feature scores. Based on our dictionary, and considering filters from the Amazon website, we categorize features into four aspects: quality, cost, service, and brand. To enhance the dictionary's diversity, we reference aspect keywords collected by Nadeem Akhter (2017).

$$Aspect = \{Quality, Expense, Service, Brand\}$$

To determine the central feature score of a product, we first consider user ratings, which range from 1 to 5. Ratings of 4 and 5 indicate positive experiences, while ratings of 1 and 2 indicate negative experiences. For ratings of 3, we use the sentiment score to evaluate positivity.

$$PositiveScores_p = \frac{\sum_{i=1}^{n} review_i(word \in Aspect)}{Numberof\,review_i ofproductp}$$

$$if ratingof\,review_i > 3 or Polarity(review_i(word \in Aspect)) > 0$$

$$NegativeScores_p = \frac{\sum_{i=1}^{n} review_{p,i}(word \in Aspect)}{Numberof\,review_{p,i}}$$

$$if ratingof\,review_i < 3 or Polarity\left(review_i(word \in Aspect)\right) < 0$$

$$AspectScores_p = PositiveScores_p + NegativeScores_p$$

where $review_{p,i}$ refers to the review of a particular product p, and review (word ∈ Aspect) indicates the identification of aspects within each word of $review_i$. We first use review ratings as a condition; when the rating is neutral, which is 3, we rely on the review's polarity score as the deciding factor. $PositiveScores_p$ and $NegativeScores_p$ compute the positive and negative scores for the four aspects of the specific product p, $AspectScores_p$ represents the final scores for these aspects.

Furthermore, we apply normalization to scale $AspectScores_p$ within the range of [0, 1], resulting in the normalized aspect score (NAS) for each product p.

$$\mathrm{NAS}_p = \frac{AspectScores_p - \min(AspectScores_p)}{\max\left(AspectScores_p\right) - \min(AspectScores_p)}$$

where $\max\left(AspectScores_p\right)$ and $\min(AspectScores_p)$ denote the maximum and minimum values of $AspectScores_p$ across all aspects.

Personalized recommendation module

Finally, the personalized recommendation module integrates the trust-weighted dataset into a hybrid recommendation algorithm that combines content-based, collaborative, and popularity-based methods (Koren et al., 2009). User preference profiles inferred from credibility-weighted reviews are matched with filtered product features to generate a personalized ranking list for each user.

This hybrid ensemble ensures that recommendations reflect authentic consumer experiences and strengthens both the accuracy and trustworthiness of the recommendation process (Paul & Nikolaev, 2021).

User preference analysis

Due to the generally high rating scores, users often struggle to determine which reviews are genuinely helpful. Additionally, each person has a different interpretation of a five-star rating. Understanding each user's preferences is crucial for developing an effective recommendation system. In this module, we will calculate user preferences based on their historical posts. We infer user preferences by counting the number of aspects to which a given sentence belongs; the more a sentence relates to a particular aspect, the more that aspect matters to the user.

$$UserPreference_u = \#wordinreview \in Aspect$$

$$UPS_u = \left\{rankAspect_1, rankAspect_2, rankAspect_3, rankAspect_4\right\}$$

where $UserPreference_u$ represents the sum of word counts in the aspect list, and UPS_u denotes the ranked aspect list indicating which aspects are most important to the user. We assign rank scores in a range from 1 to 4, reflecting the significance of these four aspects, which will guide product recommendations in the next step.

Product recommendation

In this section, we will generate a product recommendation list for users. Our recommendation system employs three methodologies: content-based filtering, collaborative filtering, and hybrid recommendations.

1. Content-based filtering: We compute the product rank score based on the product aspect score and the user aspect score. We multiply the user preference score by the product-centric feature score for each aspect and sum these values to obtain the rank score. A higher rank score indicates that the product will appear higher on the recommendation list.

$$ContentScore(User, Aspect) = \sum_{i=1}^{4} UPS(User, aspect_i) * NAS(aspect_i)$$

where *UPS* denotes the user's aspect score, *NAS* denotes the product's aspect score, and our recommendation list will consist of products with the highest rank scores.

2. Collaborative filtering: We utilize a singular value decomposition (SVD)-based collaborative filtering approach. Research has shown that SVD effectively addresses the sparsity and scalability issues common in collaborative filtering (Ardiansyah et al., 2019). We first generate a user rating matrix, which is an m × n matrix representing users' ratings for products, where m is the number of products and n is the number of users. The element Ratingi,j in the matrix represents User i's rating for Product j.

Given that only a small number of customers typically leave ratings on an e-commerce platform, and many may not rate all products purchased, we encounter sparsity issues. Most collaborative filtering methods calculate user similarity to predict scores for unrated items, employing Pearson and cosine similarity to achieve this. We leverage SVD to overcome sparse matrix problems, as SVD is a matrix decomposition technique:

$$\mathrm{SVD(M)} = \mathrm{U} \times \mathrm{S} \times \mathrm{V}^T$$

where M denotes the user rating matrix, U and V are orthogonal matrices, and S is a diagonal matrix containing singular values.

$$\mathrm{S} = \begin{bmatrix} \sigma_1 & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & \sigma_r & 0 & \cdots & 0 \\ 0 & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & 0 & \cdots & 0 \end{bmatrix}$$

The main diagonal elements $\sigma_i > 0$ for $i = 1,2,\ldots,\mathrm{r}$ represent the singular values, with $\sigma_{r+1} = \sigma_p = 0$, where $p = \min(\mathrm{m},\mathrm{n})$. By applying SVD, we fill in the sparse matrix and recommend the products with the highest scores to users.

$$CollaborativeScore = SVD\left(Rating_{m,n}\right)$$

where $Rating_{m,n}$ denotes the rating score that user m gives to product n.

3.Popularity recommendation: We rank products based on the number of reviews they have received, using this as a measure of popularity to generate a list of best-selling products.

$$PopularScore(product) = numberofreviews_p$$

where p denotes to product p. In this context, we consider the number of reviews as an indicator of product popularity.

Finally, in the hybrid recommendation approach, we normalize the scores from content-based filtering, collaborative filtering, and product popularity to a common range and combine them to produce the final recommendation list. After normalization, the final hybrid score is computed as:

$$Hybrid = ContentScore + CollaborativeScore + PopularScore$$

where $\alpha, \beta,$ andγ are set according to the experimental configuration.

Experiments

Data preparation module

To develop our spam-aware recommendation mechanism, we use Amazon product reviews as the primary data source. Because our evaluation of recommendation accuracy is based on real purchase and review behaviours within the same market period, we restrict our unlabeled dataset to 2018 in order to maintain temporal alignment between (i) the spam filtering stage and (ii) the offline recommendation testing window.

Data collection

Two Amazon review datasets are utilized:

1. Labeled dataset. We use a publicly available GitHub dataset commonly referenced in prior studies, containing 30 Amazon product categories with 350 truthful and 350 deceptive reviews per category (700 per category). The dataset provides the following information: label, rating, verified purchase flag, product category, product ID, product title, review title, and review text. Although reviewer-level identifiers are not available, this dataset is domain-matched to e-commerce product reviews and, therefore, suitable for supervised feature screening and initial classifier selection.
2. Unlabeled dataset. We adopt the large-scale Amazon electronics review corpus from Nitin Jindal and Bing Liu (Jindal & Liu, 2008; Ni et al., 2019), which includes reviewer-level identifiers, Unix timestamps, product options (e.g., color, size), and verified purchase meta-information. The original dataset spans from May 1996 to October 2018 (10.7 GB, over 21 million rows). This dataset is used for analyzing reviewer behavior, detecting spammers, scoring products, and evaluating recommendations.

Data extraction and cleaning

To reduce processing volume and align with the recommendation evaluation period, we extract reviews from January 1, 2018, to October 2, 2018. Non-English reviews, incomplete records, punctuation, and numeric characters are removed from review text fields. In the unlabelled dataset, sparse fields (images, votes) were dropped.

Dataset justification and label credibility

We selected the 2018 Amazon corpus to ensure temporal coherence between spam detection signals (content + behaviour) and the subsequent recommendation testing period. Although newer Amazon review releases (e.g., Amazon-Reviews-2023 by McAuley Lab) are available, they are unlabelled for deception and represent a different market regime, which would introduce covariate shift and weaken experimental validity.

The labeled dataset (commonly referred to as Saxena, 2018) is a publicly available GitHub corpus containing balanced truthful and deceptive Amazon product reviews across 30 categories. This dataset has been adopted in multiple peer-reviewed studies (Azam et al., 2024; Bathla et al., 2022; Salminen et al., 2022; Thilini Jayasinghe & Dassanayaka, 2025) which have validated its labeling balance and suitability for supervised fake-review detection.

To further enhance reliability, we implemented an independent expert-labeled audit of 3,316 Amazon electronics reviews (three raters, majority vote) as the ground truth for all final evaluations. This two-stage design—using Saxena (2018) for feature screening and the expert audit for validation—reduces dependence on any single labeling pipeline while preserving domain consistency and reproducibility.

Final dataset composition and balance

The final dataset used for classifier training and evaluation comprises 3316 expert-labeled Amazon electronics reviews, including 1702 non-spam (genuine) and 1614 spam reviews. This near-parity distribution ($\approx$ 1.05:1) minimizes class imbalance concerns. To confirm stability, we applied stratified fivefold cross-validation, ensuring each fold preserved the same class ratio. No artificial oversampling or undersampling was necessary, as the natural distribution

provided adequate support for both spam detection and hybrid recommendation training tasks.

Review filtering module

To examine linguistic style differences between spam and non-spam reviews, we compute the cumulative distribution functions (CDFs) for several pronoun-related variables introduced in the "Research framework and methodologies" section. The CDF visualizations allow us to compare the probability distributions of each feature across the two review categories: if the curves nearly overlap, the feature is not discriminative.

As illustrated in Figs. 3 and 4, the total counts and ratios of first-, second-, and third-person pronouns are highly similar between spam and non-spam reviews. This indicates that pronoun usage habits—often considered stylistic markers of deception—do not meaningfully differentiate review types in this dataset. Consequently, pronoun frequency is excluded from our subsequent feature-importance ranking, while other lexical and structural cues (e.g., review length and semantic similarity) provide stronger discriminative power.

After removing non-letter characters and applying stemming, each review is tokenized and its length is computed. Figure 5 illustrates the CDF distribution of review length and shows that spam reviews are substantially shorter, confirming review length as a discriminative feature.

To assess the textual resemblance among reviews, we compute a cosine similarity score for each review by weighting the labels of its ten most similar reviews with their corresponding cosine similarity values. A higher cosine similarity score indicates stronger resemblance to other reviews within the dataset.

As illustrated in Fig. 6, non-spam reviews generally exhibit higher cosine similarity scores than spam reviews, suggesting that genuine feedback tends to share more consistent linguistic and semantic patterns. The cumulative distribution functions further reveal that while the central tendency of similarity overlaps between the two groups, the upper tail (similarity > 0.85) displays a distinct

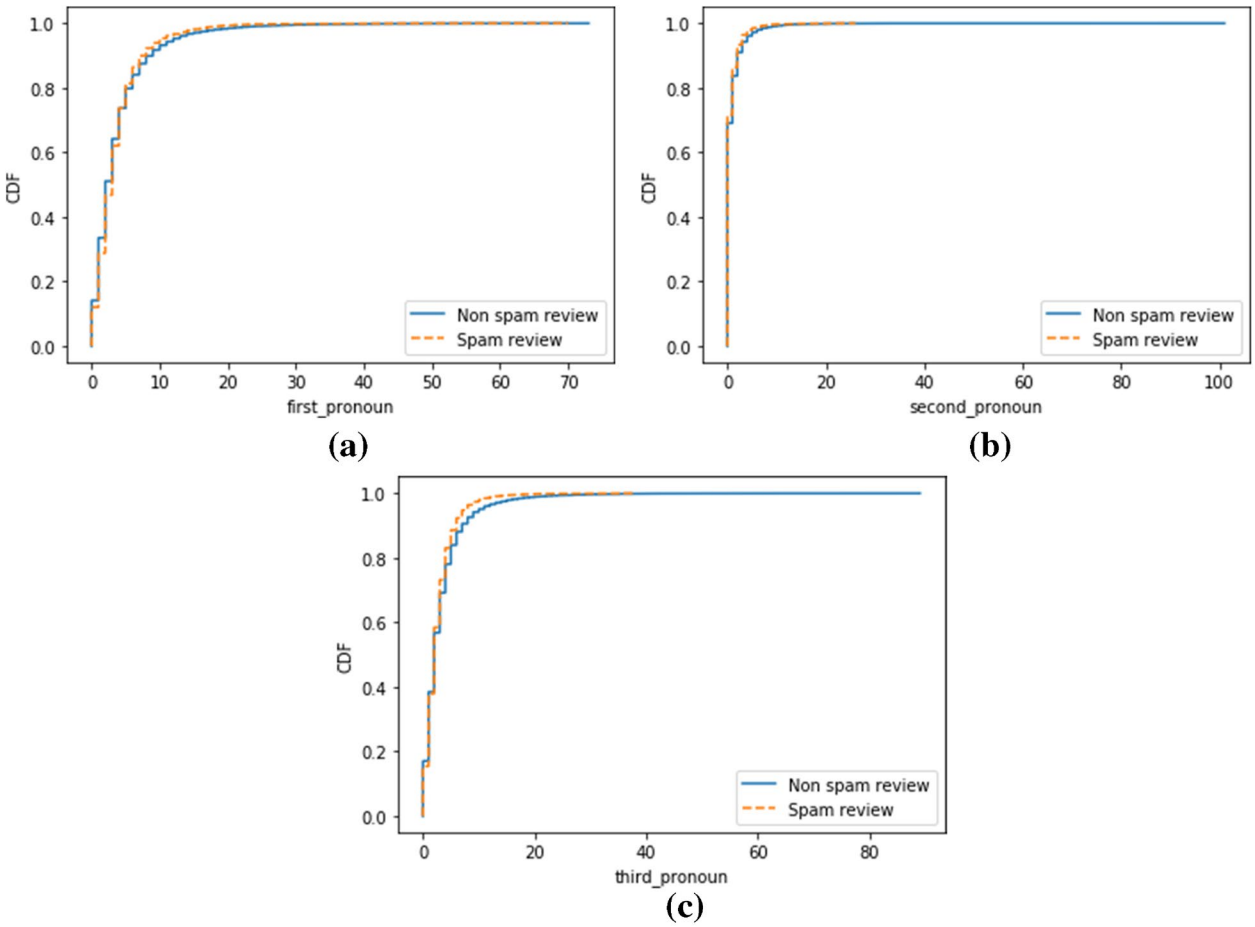


Fig. 3 **a–c** Cumulative distribution functions of number of pronouns

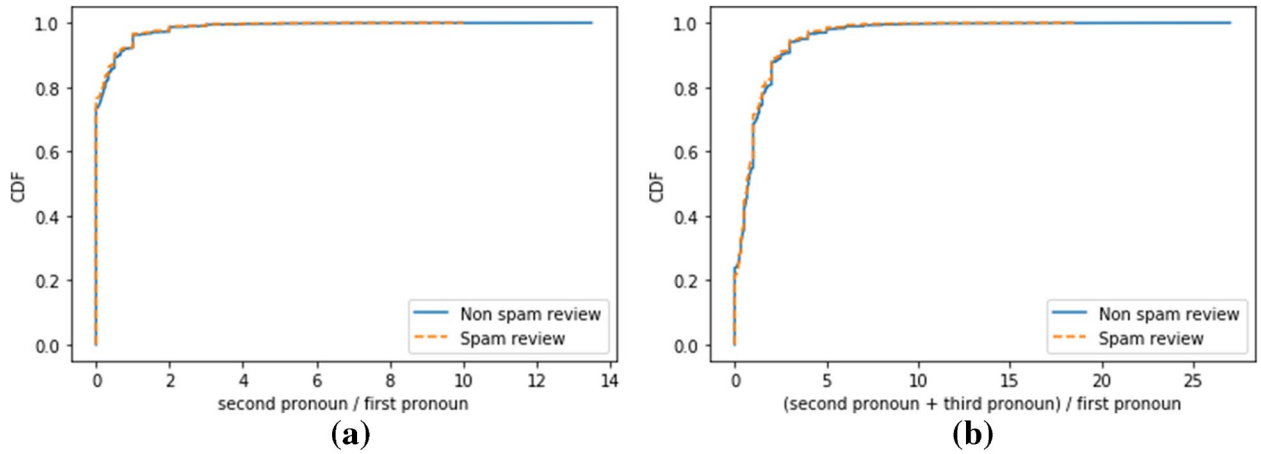


Fig. 4 **a**, **b** Cumulative distribution functions of the ratio of pronouns

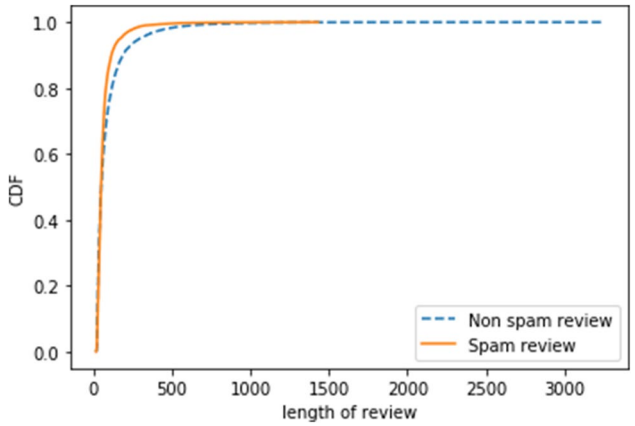


Fig. 5 Cumulative distribution functions of the review length

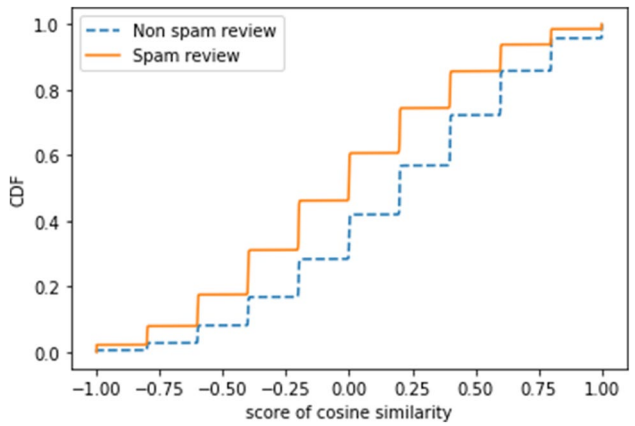


Fig. 6 Cumulative distribution functions of the cosine similarity score

rightward shift for spam reviews. This pattern indicates that deceptive reviews often exhibit clustered redundancy—consistent with coordinated generation behaviors—which validates the inclusion of semantic similarity as a discriminative feature. However, certain spam reviews closely mimic the writing style of authentic ones, making them more challenging to distinguish and underscoring the need for multi-feature integration in detection.

For the n-gram analysis, we employ the Python NLTK toolkit to generate unigram, bigram, and trigram representations for each review after text preprocessing and stemming. These representations capture different levels of contextual dependency among words.

Table 4 summarizes the comparative results after applying the n-gram transformation, showing how the inclusion of higher-order n-grams (bi- and tri-grams) enhances the model's ability to capture phrase-level semantics beyond single-word (unigram) features.

After constructing the n-gram representations, we apply TF-IDF (max features = 500) to quantify term importance and Word2Vec (vector dimension = 100) to obtain unsupervised word embeddings. TF-IDF captures discriminative term weights, whereas Word2Vec encodes distributional semantics at the phrase/sentence level. We also compute review-level cosine similarity using Word2Vec embeddings, since they provide a compact, continuous representation for each review.

We compute each review's polarity (− 1 to 1) and subjectivity (0 to 1) using TextBlob. Figure 7 a reports the CDFs of polarity and subjectivity, showing that spam reviews exhibit systematically higher polarity and subjectivity than non-spam reviews. Figure 7 b focuses on the lower tail (CDF < ~ 0.10), indicating that spam reviews are more sentiment-extreme in this region. These patterns underscore the value of sentiment features for discriminating spam.

Table 4 Example of n-gram

Raw review	The perfect keeper for my mini bose headphones. I love the individual pocket for the charger and exactly the right size
Unigram	"the," "perfect," "keeper," "for," "my," "mini," "bose," "headphones," "I," "love," "the," "individual," "pocket," "for," "the," "charger," "and," "exactly," "the," "right," "size"
Bigram	"the perfect," "perfect keeper," "keeper for," "for my," "my mini," "mini bose," "bose headphones" "headphones I," "I love," "love the," "the individual," "individual pocket," "pocket for," "for the," "the charger," "charger and," "and exactly," "exactly the," "the right," "right size"
Trigram	"the perfect keeper," "perfect keeper for," "keeper for my," "for my mini," "my mini bose," "mini bose headphones," "bose headphones I" "headphones I love," "I love the," "love the individual," "the individual pocket," "individual pocket for," "pocket for the," "for the charger," "the charger and," "charger and exactly," "and exactly the," "exactly the right," "the right size"

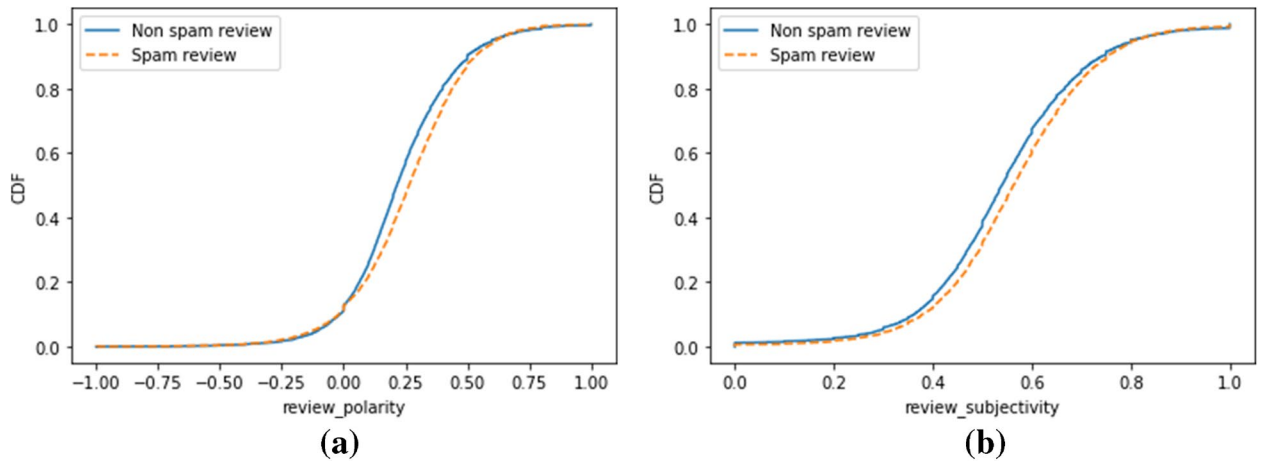


Fig. 7 **a**, **b** Cumulative distribution functions of the review polarity and subjectivity

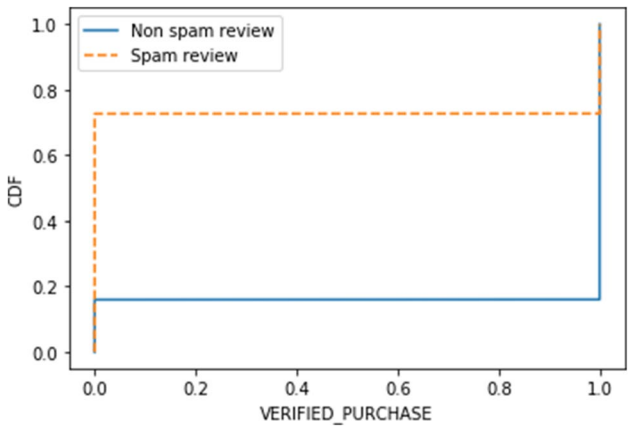


Fig. 8 Cumulative distribution functions of the verified purchased

For the metadata feature, we examine the Verified Purchase flag as a binary indicator of transaction authenticity. A value of 1 denotes that the reviewer actually purchased the product on the platform, whereas 0 indicates that the review was posted without a verified transaction.

As illustrated in Fig. 8, more than 80% of non-spam reviews are verified purchases (1), while over 70% of spam reviews are unverified (0). This pronounced separation between the two distributions highlights that verified-purchase information is a highly discriminative cue for distinguishing between authentic and deceptive reviews.

Reviewer detection module

To further distinguish spam reviews, we analyze reviewer behavior to detect potential spammers. This section details the thresholds and formulas described in the "Reviewer detection module" section.

First, we calculate the bias rate, which measures the number of reviews a user posts for the same product. In our dataset, products may have a product ID as well as product options (e.g., color, size). Therefore, we compute two metrics: (a) the number of reviews a reviewer posts for a product and (b) the number of reviews a reviewer posts for the same product option in one day. Based on Liu et al. (2017), we set the threshold at 3 reviews per product per day and 1 review per product option per day. Exceeding

Table 5 Indexes without word embedding

	Accuracy	*Precision*	*Recall*	*F1_Score*
Naive Bayes	0.588404	0.637691	0.581815	0.608473
Decision Tree	0.733549	0.733529	0.733629	0.733579
Logistic Regression	0.792867	0.799789	0.794404	0.797087
Random Forest	0.795346	0.799778	0.796571	0.797328
SVM	0.725539	0.73062	0.726963	0.728787

Table 6 Indexes of W2V

	Accuracy	*Precision*	*Recall*	*F1_Score*
Naive Bayes	0.672897	0.675916	0.671268	0.673584
Decision Tree	0.722678	0.722587	0.722653	0.72262
Logistic Regression	0.795918	0.801666	0.797314	0.799484
Random Forest	0.812893	0.817404	0.81411	**0.815754**
SVM	0.784093	0.792392	0.785793	0.789079

Table 7 Indexes of TF-IDF for Unigram

	Accuracy	*Precision*	*Recall*	*F1_Score*
Naive Bayes	0.641617	0.64143	0.641244	0.641337
Decision Tree	0.755865	0.757856	0.756704	0.75728
Logistic Regression	0.794965	0.80182	0.796491	0.799147
Random Forest	0.801831	0.804407	0.802757	0.803581
SVM	0.782567	0.79794	0.784877	0.791355

Table 8 Indexes of TF-IDF for Bigram

	Accuracy	*Precision*	*Recall*	*F1_Score*
Naive Bayes	0.609002	0.608955	0.609	0.608977
Decision Tree	0.756628	0.757911	0.757292	0.757601
Logistic Regression	0.794965	0.802035	0.796514	0.799265
Random Forest	0.806218	0.811856	0.807586	0.809715
SVM	0.782377	0.797795	0.78469	0.791188

Table 9 Indexes of TF-IDF for Trigram

	Accuracy	*Precision*	*Recall*	*F1_Score*
Naive Bayes	0.56132	0.568814	0.564082	0.566438
Decision Tree	0.757009	0.758099	0.757618	0.757858
Logistic Regression	0.794774	0.801949	0.796335	0.799132
Random Forest	0.801068	0.806473	0.802414	0.804438
SVM	0.782567	0.797832	0.784869	0.791297

Table 10 LSTM in single input and multi-input

	Accuracy	*F1-score*	*Precision*	*Recall*
Single input	0.648784	0.652468	0.657335	0.647673
Multi-input	0.779447	0.779426	0.779482	0.77937

these thresholds indicates potential manipulation aimed at diluting the review quality.

Second, we assess review burstiness by calculating the maximum number of reviews a user submits in one day. Drawing on Hussain et al. (2020) and Liu et al. (2017), we set the burst threshold at 5 reviews per day; users exceeding this threshold are flagged as abnormal.

Lastly, we evaluate reviewer content by examining review length and computing the cosine similarity among all reviews authored by the same user. Following previous research (Hussain et al., 2020; P. Liu et al., 2017), we use a threshold of 5 words for review length (UL) and 0.9 for cosine similarity (UCS). Consistently short reviews or high inter-review similarity suggest that the reviews are likely generated without genuine product experience.

These combined behavioral metrics help identify spammers, thereby improving the overall quality of the review dataset used for product recommendation.

Product scoring analysis module

This module recalculates product ratings by integrating our spam detection outcomes with aspect-based sentiment analysis. We employ both traditional machine learning and deep learning techniques to classify reviews using TF-IDF and Word2Vec features. Tables 5, 6, 7, 8, and 9 present the performance of various machine learning classifiers.

In addition, a multi-input LSTM model is used to classify reviews based on Word2Vec features combined with other non-embedding features. As Table 10 indicates, the multi-input LSTM model achieves higher accuracy and F1-scores compared to models using a single feature type, confirming that our feature selection enhances spam classification.

After evaluating model performances, we apply the best-performing classifier—random forest with Word2Vec features—to our unlabeled dataset. We then proceed with spammer behavior detection. The SpammerScore, detailed in the "Review trust score" section, aggregates five behavioral metrics; if a reviewer's score exceeds 2, their reviews are classified as spam.

For product scoring, we first construct an aspect dictionary. Adapting the dictionary from Akhtar et al. (2017) for TripAdvisor hotels, we refine the list to focus on four key aspects—quality, expense, service, and brand—which are critical for electronic products. For the brand aspect, we include only brands with sufficient representation on Amazon (e.g., more than 20 products) to ensure stable and meaningful aggregation.

To compute a product's aspect score, we compare the stemmed review text with the aspect dictionary. Using both the review rating and its sentiment score, we assign + 1 for positive reviews and − 1.3 for negative ones, reflecting the greater impact of negative reviews. The overall aspect score for a product is then calculated as the sum of its review aspect scores divided by the number of reviews. Finally, we normalize these scores to the range [0, 1] to obtain the normalized aspect score (NAS) for each product.

Personal recommendation module

After filtering out spam reviews, we partition the data into training (January 1, 2017–December 31, 2017) and testing sets (January 1, 2018–October 2, 2018) and select active users for analysis.

In the content-based approach, we first compute a reviewer's aspect scores based on the frequency with which they mention words from four predefined aspects (quality, expense, service, and brand). The more often a user mentions words from a particular aspect, the higher their preference for that aspect. We then rank these aspect scores using a Python ranking package (see Tables 11 and 12). By multiplying the reviewer's aspect scores with the corresponding product aspect scores and summing the results, we obtain an overall score for each reviewer–product pair. The product with the highest score is recommended to the reviewer, and the recommendations are evaluated using the testing data.

In the collaborative filtering approach, we construct a reviewer–product rating matrix from the available ratings. To address sparsity, we apply singular value decomposition (SVD) to fill in missing values, thus enabling prediction of ratings for all reviewer–product pairs. The product with the highest predicted rating is then recommended to each reviewer.

In the popularity approach, we rank products by the number of reviews, with higher review counts indicating greater popularity. The most popular products are recommended as a separate recommendation list.

Finally, in the hybrid method, we normalize and combine the scores from the content-based, collaborative filtering, and popularity methods to generate a comprehensive recommendation list for each reviewer.

Table 11 The original result of User Aspect Score

Reviewer id	*QualityAvg*	*ExpenseAvg*	*ServiceAvg*	*BrandAvg*
A2VODABWSVHV8E	6.0	6.0	3.0	2.0
AYRAS2X1L4WNY	9.0	1.0	5.0	5.0
AV1KOP8BCYZUJ	1.0	2.0	1.0	4.0

Table 12 The rank result of User Aspect Score

Reviewer id	*norQuality*	*norExpense*	*norService*	*norBrand*
A2VODABWSVHV8E	4.0	4.0	2.0	1.0
AYRAS2X1L4WNY	4.0	1.0	3.0	3.0
AV1KOP8BCYZUJ	2.0	3.0	2.0	4.0

Results and evaluation

Evaluation of detecting spam reviews

In this section, we present the experimental results for our spam review detection modules. First, we evaluate our feature engineering using standard metrics such as precision, recall, F1-score, and accuracy (defined as $Accuracy = \frac{TP+TN}{Total}$). Next, we compare several supervised learning algorithms—including Naïve Bayes, decision tree, logistic regression, random forest, support vector machine, and LSTM—using our engineered features. Finally, we assess the impact of integrating the reviewer detection module.

Evaluation of features

Using a random forest classifier, we measure the relative contribution of each feature. Table 13 shows the importance

Table 13 Features importance

Features	Importance
Verified Purchase	0.313628
Review length	0.132096
Cosine similarity score	0.110121
Sentiment difference between review and title	0.062471
Review subjectivity	0.056216
Review polarity	0.048539
Difference between rating and review sentiment score	0.045011
(Second pronoun + Third pronoun)/First pronoun	0.034564
First pronoun	0.030094
Difference between rating and title sentiment score	0.025913
Title subjectivity	0.025864
Title polarity	0.023677
Third pronoun	0.022384
Second pronoun/First pronoun	0.007164
Second pronoun	0.006022

ranking. Verified Purchase, review length, cosine similarity, and sentiment-related deviations contribute most to classifier performance. This provides empirical grounding for the theoretically motivated feature families in the "Spam review detection" section and their operationalization in the "Review filtering module" section.

We further compare the performance of different word embedding approaches. As shown in Table 14, Word2Vec consistently outperforms TF-IDF variants, indicating that distributed semantic representations provide stronger discriminative power. Word embeddings refer to numerical representations of words in a multi-dimensional vector space, allowing similar words to have closer representations based on context.

Evaluation of reviewer detection

To enhance spam detection accuracy, we integrated a reviewer detection module that analyzes reviewer behavior. We randomly selected 3316 reviews from our unlabeled dataset and obtained expert annotations (following Algur and Biradar (2015)) to serve as ground truth. We compared three configurations: (1) using only the review detection module, (2) adding the reviewer detection module with a spammer score threshold of 2, and (3) adding the reviewer detection module with a threshold of 3.

Our results (see Table 15) indicate that setting the spammer score threshold to 2 yields the best performance, with the highest accuracy and F1-score. Increasing the threshold to 3 did not significantly improve performance, as it underestimates the number of spammers.

These findings confirm that incorporating reviewer behavior with an appropriate threshold improves the overall spam review detection accuracy.

Table 14 Word embedding methods evaluation

Word embedding method	Accuracy	Precision	Recall	F1-score
Without word embedding	0.795346	0.799778	0.796571	0.797328
Word2Vec	0.812893	0.817404	0.81411	**0.815754**
Unigram of tf-idf	0.801831	0.804407	0.802757	0.803581
Bigram of tf-idf	0.806218	0.811856	0.807586	0.809715
Trigram of tf-idf	0.801068	0.806473	0.802414	0.804438

Evaluation of recommendation accuracy

In this section, we evaluate the accuracy of our product recommendation system. The recommendations are generated using an unlabeled review dataset after filtering out spam reviews. We assess recommendation accuracy based on whether a user actually purchases a recommended product and leaves a review within the testing period.

In line with established recommender-system methodologies (Ekstrand et al., 2011; Heinrich et al., 2021; J. Kim et al., 2021), we use actual historical purchases as observable signals of revealed preference. While this measure does not guarantee ex post satisfaction, it reflects ex ante consumer choice under the information set available at the time of decision, which aligns with our objective of enhancing recommendation credibility and trust prior to purchase.

Post-consumption feedback—such as returns, rating reversals, or longitudinal satisfaction drift—is typically unavailable in public review datasets. Hence, we treat realized purchases as a pragmatic and consistent proxy for evaluating the accuracy of recommendations.

Performance comparison of different recommendation approaches

We compare the performance of various recommendation methods, including content-based, collaborative, popularity-based, hybrid, and baseline random recommendation. Figure 9 presents the results, where N represents the number of products in the recommended list. The values of N are based on previous research on e-commerce recommendation systems (R. K.-W. Lee et al., 2020).

The results indicate that collaborative filtering and hybrid recommendation methods outperform others, with the hybrid approach—which integrates content-based, collaborative, and popularity-based methods—achieving the best overall performance.

Impact of spam review detection on recommendation accuracy

We further evaluate the accuracy of the hybrid recommendation system before and after filtering spam reviews. Table 16 demonstrates that removing spam reviews significantly improves recommendation accuracy across all values of N.

Table 15 Performance comparison of reviewer detection configurations

	Accuracy	**Precision**	**Recall**	**F1-score**
Review detection only	0.631484	0.399618	0.69129	0.506462
Review + reviewer detection (threshold = 2)	0.88842	0.939444	0.632856	**0.756258**
Review + reviewer detection (threshold = 3)	0.642642	0.409269	0.69129	0.514145

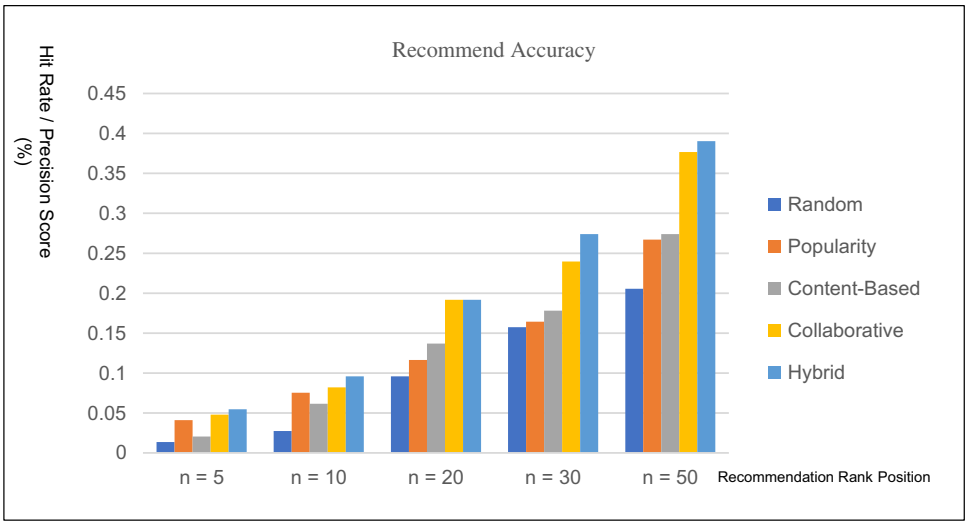


Fig. 9 Evaluation of recommendation with different Aapproaches

Table 16 Recommendation accuracy before and after removing spam reviews

Approach	$N=5$	$N=10$	$N=20$	$N=30$	$N=50$
Before Removing Spam Reviews	0.02355	0.04733	0.08368	0.11774	0.16072
After Removing Spam Reviews	0.04795	0.09589	0.19178	0.26712	0.39041

The results clearly indicate that the hybrid spam-aware model achieves consistently higher accuracy at all evaluation levels. The improvement becomes more pronounced as N increases, confirming that removing spam reviews enhances both the quality and relevance of product recommendations.

Discussion and conclusion

Research summary

The rapid expansion of e-commerce has led to an unprecedented surge in user-generated content, which profoundly influences consumer decision-making. Online review platforms offer customers valuable insights, yet they also create challenges such as information overload and reduced credibility due to spam reviews. In response, this study introduces a novel recommendation mechanism that integrates advanced spam review detection with personalized product recommendations.

Our approach leverages natural language processing, sentiment analysis, and metadata features to identify and filter out fake reviews, while also analyzing reviewer behavior to detect spammers. By using refined, trustworthy review data, our system generates product recommendations based on review credibility, user preferences, product popularity, and similarity measures. Experimental results, validated on a manually labeled dataset, demonstrate that incorporating reviewer behavior analysis significantly improves spam detection—particularly when using a random forest classifier—and that our hybrid recommendation model outperforms traditional approaches.

Discussion and research contributions

This study contributes to the literature by theoretically and empirically integrating spam review detection with personalized recommendation, two research streams that have largely evolved in parallel. Prior work on spam review detection has predominantly focused on improving classification accuracy (e.g., Jindal & Liu, 2008; Ott et al., 2013; Wu et al., 2020), typically treating spam identification as a terminal analytical task rather than as an input to downstream personalization or ranking mechanisms. In contrast, recommender systems research has commonly assumed that all user-generated content constitutes equally reliable preference signals (Koren et al., 2009). Our findings challenge this implicit assumption and show that review credibility plays a critical role in shaping personalization outcomes under realistic platform conditions.

From a spam review detection perspective, the results validate and extend earlier findings emphasizing the importance of metadata and behavioral cues, such as verified purchase indicators, posting burstiness, and cross-item similarity (Cresci et al., 2015; Mukherjee et al., 2013; Z. Wu et al., 2024). Consistent with deception and persuasion theories, the feature importance analysis confirms that deceptive reviews tend to exhibit exaggerated sentiment, reduced experiential grounding, and

repetitive linguistic patterns (Mayzlin et al., 2014; Ott et al., 2013). Importantly, our results further show that reviewer-level behavioral signals significantly enhance detection performance when combined with content-based features, supporting recent arguments that coordinated or semi-automated spam campaigns cannot be reliably identified through text-only analysis (Hou et al., 2025; Salminen et al., 2022).

From a credibility and trust signaling perspective, this study advances existing research by reconceptualizing review credibility as a continuous and probabilistic construct rather than a binary attribute. While prior studies have examined heuristic trust cues such as helpful votes or verified purchase badges (Chevalier & Mayzlin, 2006; Siering et al., 2018), our findings demonstrate that such interface-level signals are insufficient when treated as ground-truth indicators of reliability. By operationalizing both review-level and reviewer-level suspiciousness as continuous credibility scores, the proposed framework aligns with contemporary theories of algorithmic trust that emphasize graded assessments of information quality and veracity (Dong et al., 2022; Yu et al., 2022).

Most critically, this study contributes to recommender systems research by demonstrating how credibility-weighted personalization improves recommendation accuracy under realistic spam contamination. Whereas prior trust-aware recommender models often rely on tightly coupled or jointly optimized architectures (Dong et al., 2022; Guo et al., 2025), our results demonstrate that credibility can be operationalized as an external, continuous weighting mechanism that reshapes recommendation outcomes without modifying the core recommender architecture. By injecting credibility weights into existing hybrid recommendation pipelines, the framework reduces the influence of manipulated content without requiring architectural redesign. Empirically, the observed improvement in Top-N recommendation accuracy after spam filtering and weighting supports theoretical models of personalization under noisy inputs, highlighting how unfiltered manipulation can distort collaborative signals and amplify popularity bias (Jin et al., 2023).

Taken together, these findings position the present study as a conceptual bridge between deception detection and personalized recommendation. The results demonstrate that credibility signals are not merely diagnostic tools for identifying spam, but actionable inputs that meaningfully reshape algorithmic decision support in digital marketplaces.

Research limitations

Despite the promising results, several limitations should be acknowledged. First, the dataset is exclusively derived from Amazon, which may limit the generalizability of our findings to other e-commerce platforms (e.g., Walmart or eBay) where review dynamics and user behaviors differ. In addition, the current analysis focuses only on English-language reviews; extending the framework to multilingual settings would require further adaptation.

Second, the labeled dataset used for spam review detection lacks rich reviewer attributes (e.g., detailed profiles and long-term behavioral histories), which may constrain the effectiveness of reviewer-level credibility estimation. Third, the proposed framework assumes that each user and product has at least one review, which limits applicability under cold-start conditions or for passive users who rarely generate textual feedback. Future extensions may incorporate implicit behavioral traces—such as browsing logs, dwell time, and clickstream signals—as complementary indicators of user preference and credibility when review text is unavailable.

Fourth, our approach relies on supervised learning and thus requires high-quality labeled data. The scarcity of reliable labeled datasets across domains and platforms may affect robustness and scalability. Moreover, our evaluation relies on revealed preferences proxied by purchase outcomes, which reflect decision-stage behavior rather than post-consumption utility. Due to the limited availability of satisfaction, return, or complaint data in open datasets, we could not assess recommendation effectiveness at the post-purchase stage.

Finally, although our framework incorporates reviewer-behavior signals that are less sensitive to linguistic manipulation, the current feature design is primarily calibrated on human-authored opinion spam. Given the growing prevalence of AI-generated reviews, future research should explicitly evaluate temporal robustness and portability using newer datasets (e.g., Amazon Reviews 2023) and incorporate AI-aware detection modules to address emerging generative deception.

Future work

Future research can extend this study in several directions. First, incorporating datasets from multiple platforms and product categories would improve external validity and provide deeper insights into platform-specific manipulation strategies. Second, extending the framework to multilingual and multimodal reviews (e.g., images and videos) may further enhance both detection performance and recommendation quality.

Third, richer behavioral and relational data—such as browsing histories, dwell time, clickstream patterns, and social-network relationships—could improve the identification of coordinated manipulation and strengthen credibility-weighted personalization. In addition, adopting semi-supervised or weakly supervised learning strategies may reduce dependence on scarce labeled datasets and improve scalability in fast-evolving online environments.

Finally, future work may examine cross-cultural contexts and product types (e.g., experiential vs. utilitarian goods), as these factors may shape review-writing norms, deception

strategies, and the effectiveness of credibility-aware recommendation mechanisms. Such extensions would further clarify the boundary conditions under which spam-aware personalization provides the greatest value.

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Data Availability The datasets used in this study are publicly available. The labeled dataset for spam review detection is accessible at: https://github.com/aayush210789/Deception-Detection-on-Amazon-reviews-dataset. The unlabeled Amazon review datasets are derived from publicly available corpora as described in Jindal and Liu (2008) and Ni et al. (2019). Further details are available from the corresponding author upon reasonable request.

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